higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

9 April 2025 (X-paper)
09:00–12:00

Drawing instruments may be used.
Non-programmable calculators may NOT be used.

This question paper consists of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only a black or blue pen.
5. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 The threshold voltage can be defined as:

A The voltage difference between logic levels 1 & 0.
B The minimum and maximum voltage levels which determine the logic.
C The single voltage output of a logic gate.
D The difference voltage specification for different logic gates.

1.1.2 The decimal equivalent for FA, C16 is:

A $255, 75_{10}$
B $250, 75_{10}$
C $128,25_{10}$
D $235, 25_{10}$

1.1.3 Which ONE of the following numbers is not a valid number system?

A 802_{10}
B 101_{16}
C 672_{8}
D 801_{8}

1.1.4 What is the simplest form of the Boolean expression $A + \bar{A}B$?

A A
B A + B
C 1
D 0

1.1.5 The two's complement of 10101_2 is:

A 01100
B 01011
C 01010
D 01001

(5 × 1) (5)

1.2 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1 All flip-flops in an asynchronous counter change state at the same time.

1.2.2 Shift registers are used to store and transfer data.

1.2.3 A half-adder is normally used when a carry-input may be applied.

1.2.4 When J=K=1, a JK-flip-flop is equivalent to a T-type flip-flop.

1.2.5 De-Morgan's theorem states that $\overline{A+B} = \bar{A}.\bar{B}$.

(5 × 1) (5)
[10]

QUESTION 2

2.1. Do the following calculations in the binary system. Show all calculations and conversions.

2.1.1 $11{,}25_{10}$ - $16{,}4_{8}$

Use the two's-complement method to convert the final answer to a decimal number.

2.1.2 $26{,}3_{8} \div 9, 5_{10}$

Convert the final answer to the octal number system.

2.1.3 $D{,}6_{16} \times 3, 2_{8}$

Convert the final answer to the hexadecimal number system.

(3 × 6) (18)

2.2 Write $84{,}5_{10}$ in the following BCD codes:

2.2.1 Excess-3 code

2.2.2 2421 code

(2 × 3) (6)
[24]

QUESTION 3

3.1 Write the Boolean expression for the following logic gate showing how the output (Q) algebraically relates to the inputs (A and B):

3.1.1 NAND

3.1.2 OR

3.1.3 XNOR

3.1.4 NOT

3.1.5 AND (1 × 5) (5)

3.2 Draw the logic circuit diagram of each of the following Boolean expressions: Do not change or simplify the expression.

3.2.1 $\overline{A+B}+C)(AB+\overline{AB})=F$ (6)

3.2.2 $\overleftarrow{\overline{AB+\bar{A}B\bar{C}}+\bar{B}C}=F$ (5)

3.3 Simplify the following expressions using Boolean algebra:

3.3.1 $Z(\bar{X}+\bar{Y})+\bar{X}(Y+\bar{Z})+\bar{X}Y=F$

3.3.2 $\bar{A}BC+A\bar{B}C+\bar{A}B\bar{C}+\bar{A}\bar{B}C=F$

(2 × 5) (10)
[26]

QUESTION 4

4.1 Draw a neat diagram of an SR master slave flip-flop that uses NAND gates only. Show all input and output. (10)

4.2 Draw a neat diagram of a Module 5 up counter and provide its truth table. (10)
[20]

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QUESTION 5

5.1 Draw a neat circuit diagram of a 4 × 1 line multiplexer. (7)

5.2 Calculate the output voltage of a 4 bits D/A converter if the binary input is 1110_2 and the supply voltage is 24 V. (3)

5.3 Draw a neat block diagram of a digital voltmeter (ramp type) A/D converter. Also show the ramp voltage generated. (10)
[20]



TOTAL: 100

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